Logging and experiments management

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Recap

- Sources of data
- Data storages
  - small data
  - big data
- Data engineering
  - ETL and ELT
  - datahub
- Experiments parametrization
  - Hydra
  - Vault

Outline

- Logging
    - MLflow
- Visualization
- Frameworks for training
    - Lightning
    - Hugging face

Logging

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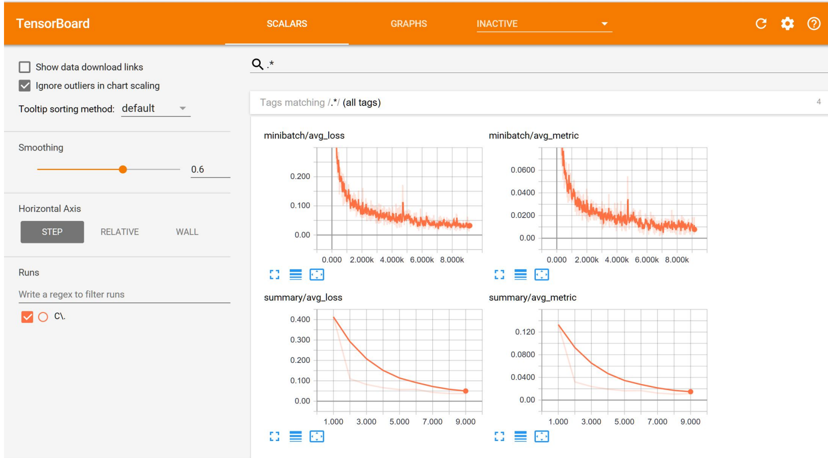
01

Experiments logging

Becomes more important with more experiments you conduct

Tools here applicable for experiments, but not production monitoring

- absence
- print
- local service
  - tensorboard
- remote service
  - MLflow
  - clear-ml
  - kubeflow
  - w&b, neptune and other proprietary



Requirements for logging

- Structured records
- Comparison of experiments
- Saving artifacts
- Distributed storage
- Realtime writing
- Realtime visualization

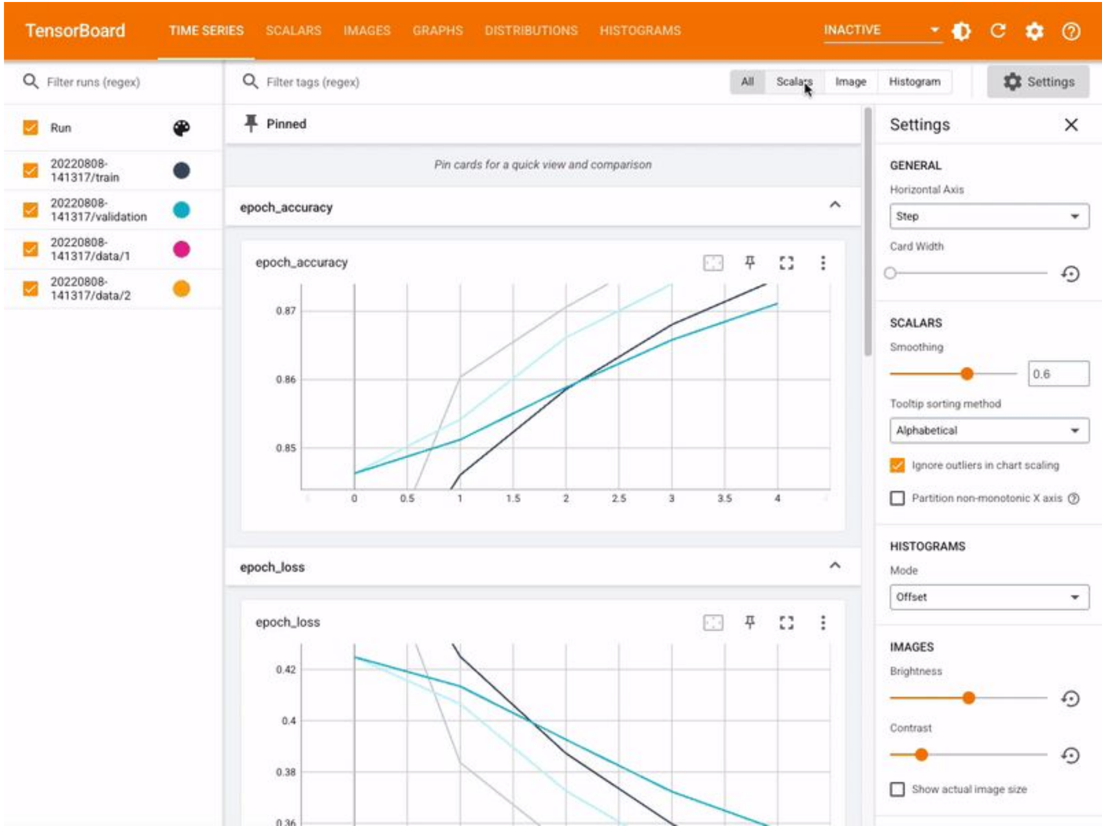
More logging

- Sacred
- Flyte
- Comet
- Valohai
- Polyaxon
- Domino Data Lab
- Cortex
- ....

See overviews: one, two

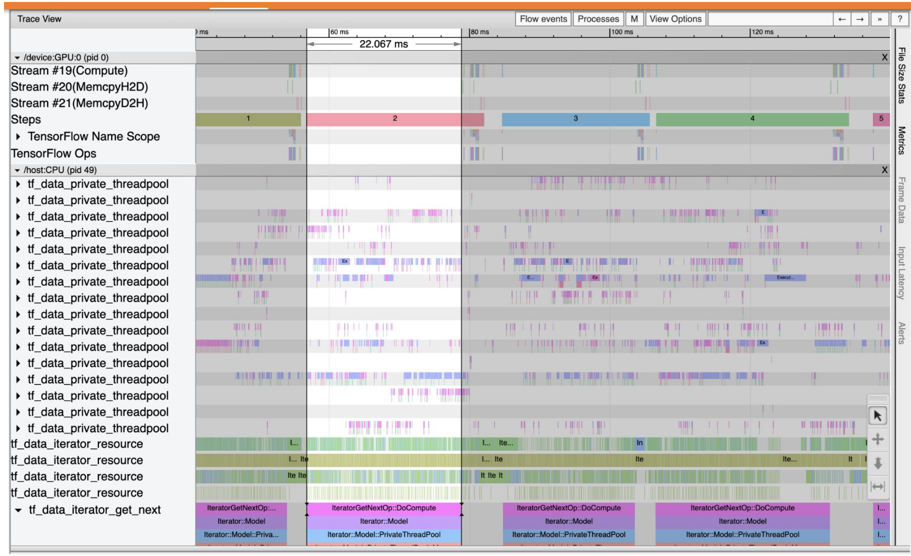
Tensorboard

- Is part of Tensorflow ecosystem but delivered as separate package
- Released in 2017
- Better suited to local running
- Authors wanted to extend it to full fledged solution but didn't succeed (see TensorBoard.dev)



Capabilities of Tensorboard

- Tracking and visualizing metrics such as loss and accuracy
- Visualizing the model graph (ops and layers)
- Viewing histograms of weights, biases, or other tensors as they change over time
- Projecting embeddings to a lower dimensional space
- Displaying images, text, and audio data
- Profiling TensorFlow programs



MLflow

- Tracking
  - Record and query experiments: code, data, config, and results
- Projects
  - Package data science code in a format to reproduce runs on any platform
- Models
  - Deploy machine learning models in diverse serving environments
- Model Registry
  - Store, annotate, discover, and manage models in a central repository

https://mlflow.org/docs/latest/index.html

Inference services

Simple deployment tool suitable for relatively small projects.

Alternatives:

- MLServer (part of Seldon)
- Triton inference server (covered later in this course)

API for inference:

https://github.com/kserve/open-inference-protocol

ClearML

https://clear.ml/

- ClearML DataOps
- ClearML Hyper-Datasets
- ClearML Experiment
- ClearML Train
- ClearML Reports
- ClearML Modelstore
- ClearML Pipelines
- ClearML Deploy
- ClearML Orchestrate
- ClearML Schedule
- ClearML Compute
- ClearML Autoscalers
- ClearML Enterprise

Kubeflow

ML suite for Kubernetes (includes tracking)

W&B, neptune
и прочая проприетарщина

- neptune.ai
  - неплохие обзоры рынка и база знаний
- Weights and biases
  - предоставляют self-hosted версию

Visualisation

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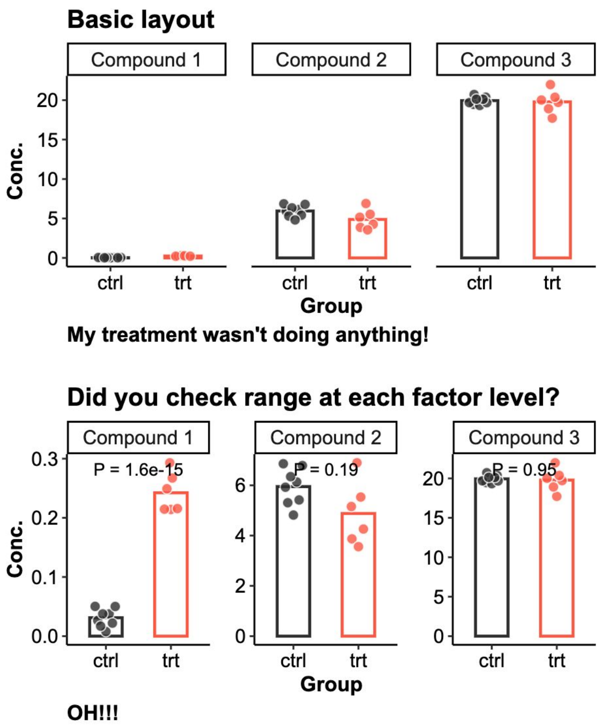
Visualisations

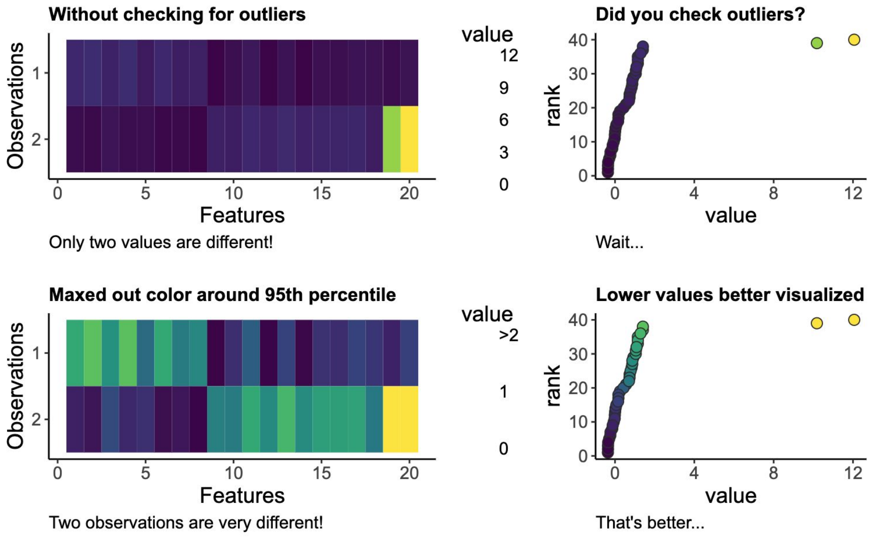
Comprehensive list of visualisations:

https://datavizcatalogue.com/

Bad cases of visualization

https://github.com/cxli233/FriendsDontLetFriends



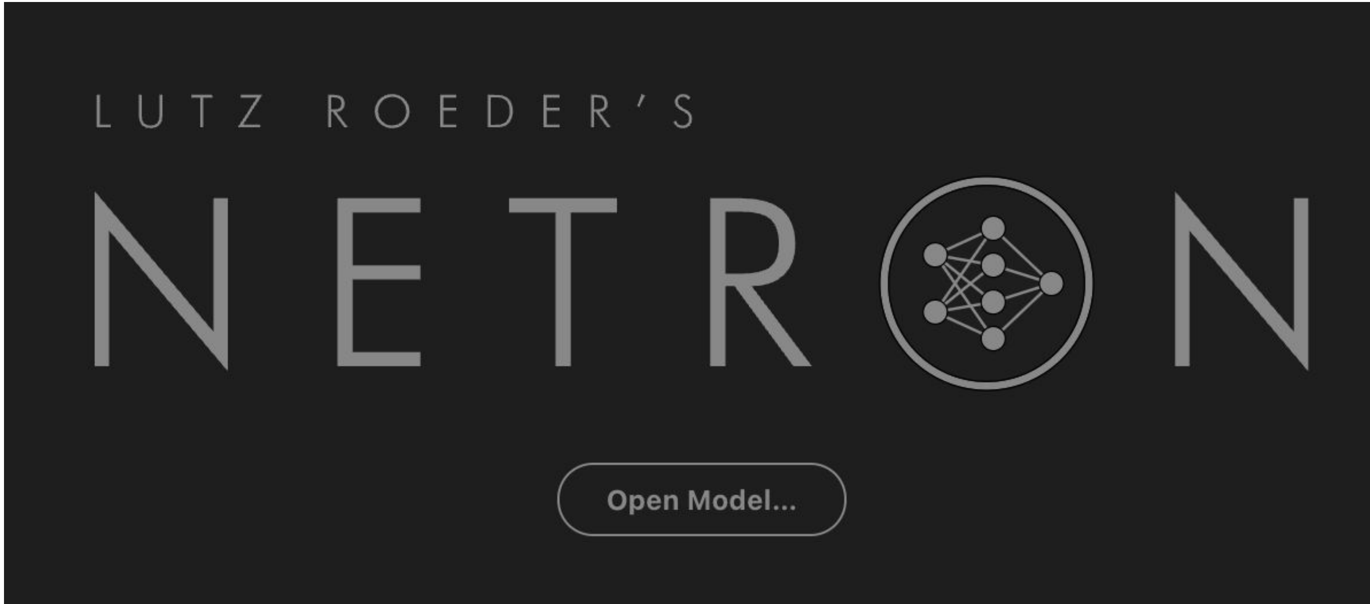


Netron app

Way to visualize your DL models

https://github.com/lutzroeder/netron

https://netron.app/



Frameworks for training

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Experiments Runners

- Pytorch
    - Lightning
    - Hugging face (also supports TF)
    - Ignite (stagnates)
    - Catalyst (already dead)
- Ray
- DVC Experiments
- MLFlow Experiments
- pachyderm
- fast.ai (*very* poor coding level e.g. import *)
- ....

P.S. see https://pytorch.org/ecosystem/

Profiling in Pytorch

- https://pytorch.org/tutorials/recipes/recipes/profiler_recipe.html
- https://pytorch.org/tutorials/beginner/profiler.html
- https://pytorch.org/docs/stable/profiler.html

```
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                             Name      Self CPU     CPU total  CPU time avg    # of Calls
---------------------------------  ------------  ------------  ------------  ------------
                  model_inference       5.509ms      57.503ms      57.503ms             1
                     aten::conv2d     231.000us      31.931ms       1.597ms            20
                aten::convolution     250.000us      31.700ms       1.585ms            20
               aten::_convolution     336.000us      31.450ms       1.573ms            20
         aten::mkldnn_convolution      30.838ms      31.114ms       1.556ms            20
                 aten::batch_norm     211.000us      14.693ms     734.650us            20
     aten::_batch_norm_impl_index     319.000us      14.482ms     724.100us            20
          aten::native_batch_norm       9.229ms      14.109ms     705.450us            20
                       aten::mean     332.000us       2.631ms     125.286us            21
                     aten::select       1.668ms       2.292ms       8.988us           255
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Self CPU time total: 57.549m
```

Revise

- Logging
  - MLflow
- Visualization
- Frameworks for training
  - Lightning
  - Hugging face

Thanks for attention!

Questions?

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